

RareDXAI: A Human Phenotype Ontology-Based Framework for Rare Disease Prediction

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1. Abstract

Background: Rare genetic diseases affect more than 300 million people worldwide. Diagnosing these conditions is challenging because many distinct genetic disorders present with overlapping clinical features, such as developmental delays, speech impairments, and intellectual disability. In pediatric genetics, syndromic neurodevelopmental conditions like KBG syndrome, White-Sutton syndrome, and Xia-Gibbs syndrome share non-specific clinical signs that frequently lead to prolonged diagnostic delays lasting five to seven years.

Methods: We developed **RareDXAI**, a computational decision-support framework that uses standardized Human Phenotype Ontology (HPO) terms to perform phenotype-based rare disease classification and candidate prioritization. We curated an audited, literature-derived cohort of 385 patients with confirmed pathogenic variants across 48 peer-reviewed publications (from 52 candidate literature sources evaluated during curation, with four secondary, overlapping, or re-reported sources containing 59 candidate records quarantined under a predefined exclusion protocol). Clinical features were encoded into 78 binary HPO terms and 3 one-hot sex indicators (81 total predictor dimensions). To reduce the risk of phenotype-profile leakage, identical phenotypic profiles were grouped before applying a 60/20/20 train/validation/test split (N=230 training, N=77 validation, N=78 held-out test), and feature vocabularies were constructed strictly from training data. We trained a Random Forest model with Platt probability scaling (5-fold internal cross-calibration) and explained predictions using TreeSHAP.

Results: On the held-out test set (N=78), RareDXAI correctly classified 77 out of 78 patients, corresponding to a held-out classification accuracy of **98.72%** (Wilson 95% CI: 93.09%–99.77%), balanced accuracy of **96.30%**, macro precision of **99.45%**, macro recall of **96.30%**, macro specificity of **98.15%**, and macro F1-score of **97.76%**. Multi-class threshold-agnostic evaluation on this held-out test set yielded a macro AUROC of **1.0000** and macro AUPRC of **1.0000**, with a multiclass Brier score of **0.0419** and Expected Calibration Error (ECE) of **8.59%**. Five-fold stratified grouped cross-validation demonstrated high consistency (97.14% $\pm$ 2.23% accuracy; macro F1: 94.89% $\pm$ 4.04%). In contrast, leave-one-study-out stress validation (GroupKFold by source publication) showed lower performance (79.62% $\pm$ 27.05% accuracy; macro F1: 59.51% $\pm$ 26.81%), indicating that publication-specific phenotype reporting patterns affect model transferability. TreeSHAP analysis identified standardized HPO features strongly associated with model decision boundaries, such as macrodontia (HP:0001572) for KBG syndrome, autism spectrum traits (HP:0000717) for White-Sutton syndrome, and thin upper lip vermilion (HP:0000219) for Xia-Gibbs syndrome. Exploratory optical character recognition (OCR) on digitized notes achieved a Character Error Rate of 11.42% and Word Error Rate of 92.00%.

Conclusions: RareDXAI provides an interpretable phenotype-based classification framework and a reproducible baseline for rare disease decision support. While held-out test classification performance is high under profile-grouped evaluation, the substantial performance drop in source-grouped stress testing highlights that independent external and prospective clinical validation remain necessary before deployment.

Keywords: Rare Diseases, Human Phenotype Ontology, Machine Learning, Random Forest, Model Interpretability, SHAP, Clinical Decision Support, Neurodevelopmental Disorders.

2. Introduction

Rare diseases are medical conditions that affect a small fraction of the population, typically defined as fewer than 1 in 2,000 individuals in Europe or fewer than 200,000 individuals in the United States (Nguengang Wakap et al., 2020; Boycott et al., 2018). Although each individual disorder is rare, there are over 7,000 recognized rare genetic conditions, which together affect an estimated 300 to 400 million people globally.

Most rare diseases have an underlying genetic cause and manifest during infancy or early childhood. Despite advances in high-throughput DNA sequencing—including whole-exome sequencing (WES) and whole-genome sequencing (WGS)—affected patients and their families frequently experience a prolonged and stressful diagnostic odyssey. On average, obtaining a correct diagnosis requires five to seven years, multiple specialist visits, and several initial misdiagnoses.

A principal contributor to this diagnostic delay is clinical phenotypic overlap. Many genetic conditions share broad, non-specific neurodevelopmental symptoms, such as intellectual disability, motor milestone delays, speech impairments, and behavioral challenges. This diagnostic challenge is particularly pronounced among three syndromic neurodevelopmental conditions:

1. KBG Syndrome (MIM #148050): Caused by heterozygous loss-of-function mutations or microdeletions in the *ANKRD11* gene on chromosome 16q24.3 (Martinez-Cayuelas et al., 2023; Sirmaci et al., 2011). Characteristic clinical findings include unusually large upper front teeth (macrodonia of the central incisors), a triangular facial appearance, prominent eyebrows, short stature, skeletal hand differences (such as brachydactyly or short fifth fingers), and mild-to-moderate intellectual disability.

2. White-Sutton Syndrome (MIM #616364): Caused by heterozygous *de novo* mutations in the *POGZ* gene on chromosome 1q21.3 (Assia Batzir et al., 2020; White et al., 2016). Common clinical features include developmental delay, intellectual disability, speech impairment, autism spectrum disorder traits, small head size (microcephaly), and distinctive facial features.

3. Xia-Gibbs Syndrome (MIM #615829): Caused by heterozygous *de novo* truncating mutations in the *AHDC1* gene on chromosome 1p36.11 (Khayat et al., 2021; Xia et al., 2014). Hallmark signs include severe infantile hypotonia (low muscle tone), global developmental delay, expressive speech absence or delay, a broad forehead, downward-slanting palpebral fissures, a thin upper lip vermilion, structural brain anomalies, and sleep disturbances such as obstructive sleep apnea.

Because these three syndromes share non-specific neurodevelopmental manifestations, differentiating among them based purely on routine initial clinical evaluation is difficult. The current model is a closed-set three-class classification system focusing on these three conditions and should not be interpreted as a general-purpose rare disease diagnostic model.

To standardize clinical descriptions and make them computable, the biomedical community established the Human Phenotype Ontology (HPO) (Köhler et al., 2021; Gargano et al., 2024). The HPO provides a controlled vocabulary of standardized medical terms organized in a directed acyclic graph. For example, instead of recording free-text phrases like 'large central teeth,' clinicians and computational pipelines use the standardized concept HP:0001572 (Macrodonia of central incisors). This structured vocabulary allows machine-learning algorithms to analyze patient phenotypic profiles systematically.

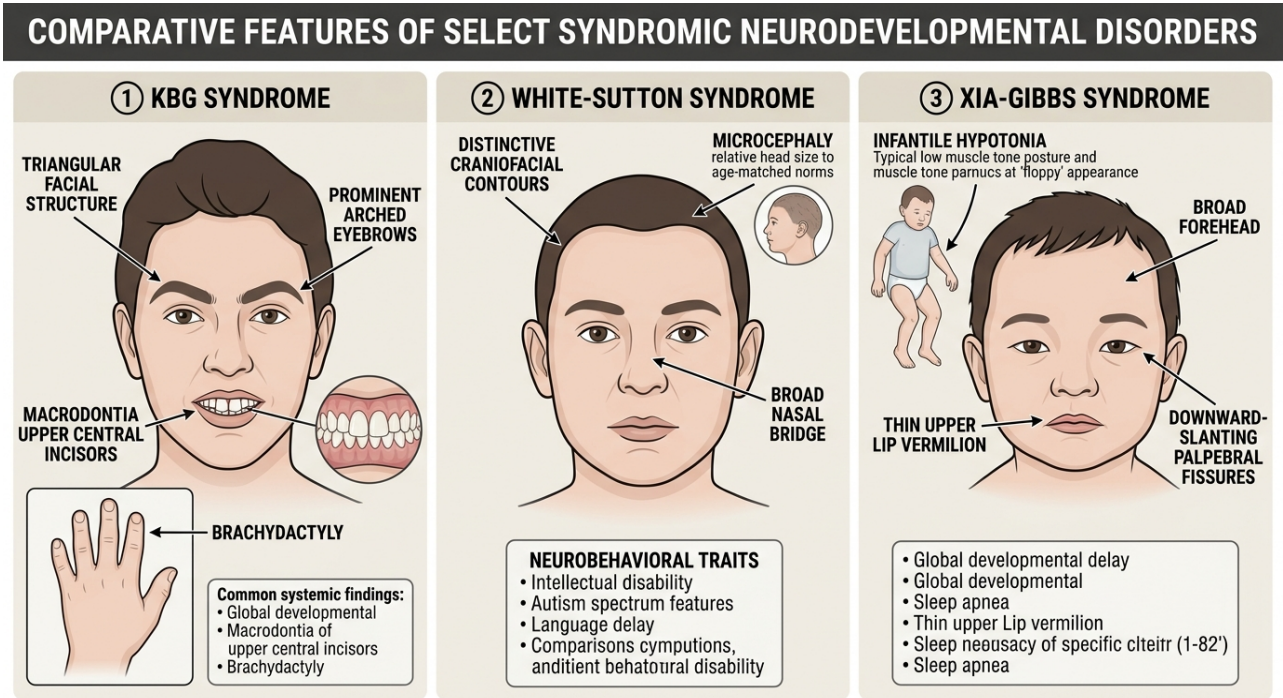


Figure 1. Representative clinical and phenotypic features associated with KBG, White-Sutton, and Xia-Gibbs syndromes. The illustration is intended for visual context and does not represent all affected individuals.

3. Literature Survey

Computational methods for rare disease analysis have advanced from manual expert tables to ontology-driven machine learning, natural language processing (NLP), and deep learning. Mining individual patient case reports from published literature and converting them into structured, computable datasets is a well-established practice in computational genetics.

We reviewed 15 peer-reviewed studies published between 2020 and 2026 that directly inform the design of RareDXAI. The Phenopacket Store (Ladewig et al., 2023) provides the closest methodological precedent to our study, demonstrating that structured extraction of published case reports enables reproducible benchmarking. RareDXAI builds upon this precedent by focusing on a specific three-syndrome classification problem with binary feature encoding, probability calibration, and SHAP interpretability.

Table 1. Literature Comparison Matrix (15 Peer-Reviewed Studies, 2020–2026)

Year	Authors	Title & Venue	Dataset / Source	Phenotype Format	Method	Relevance to RareDXAI
2022	Jacobsen et al.	GA4GH Phenopacket schema (Nature Biotechnology)	GA4GH clinical repositories	Hierarchical HPO + GA4GH	ISO Standard (ISO 4454:2022)	Conceptual basis for computable patient phenotyping
2023	Ladewig et al.	Phenopacket Store (Database)	Case reports from published literature	Curated patient-level HPO	Curated case repository & validation	Establishes methodological precedent; RareDXAI adds focus on
2024	Gargano et al.	HPO in 2024 (Nucleic Acids Res)	HPO International Consortium	Controlled vocabulary (>16,000 terms)	Knowledge graph structures	Source of canonical terminology (Release 2026-06-23)
2021	Köhler et al.	HPO in 2021 (Nucleic Acids Res)	Rare disease clinical databases	Directed acyclic graph of HPO	Semantic similarity algorithms	Baseline ontology structure for feature mapping
2020	Robinson et al.	LIRICAL (Am J Hum Genet)	Real clinical cases & simulations	Observed & excluded HPO	Likelihood ratio & Bayesian inference	Supports interpretable decision support framework
2020	Birgmeier et al.	AMELIE (Sci Transl Med)	138,000+ primary literature papers	Concept extraction	Supervised machine learning	Confirms viability of literature-derived patient matching
2021	Feng et al.	PhenoTagger (Bioinformatics)	PubMed Central text corpora	Standardized HPO concept	Hybrid CNN + dictionary indexing	Explores text-to-HPO concept mapping strategies
2020	Liu et al.	Doc2HPO (BMC Bioinformatics)	Unstructured clinical notes	Interactive HPO term mapping	String parsing + NER	Upstream precedent for clinical concept parsing
2020	Zhao et al.	Phen2Gene (Nucleic Acids Res)	OMIM, Orphanet, HPO annotations	Weighted HPO disease-gene	Information theoretic gene scoring	Strong HPO term weighting akin to feature vectorization
2022	Hsieh et al.	GestaltMatcher (Nat Genet)	Patient clinical photographs	Deep facial dysmorphic embeddings	Deep convolutional Neural Networks	Neural modality for syndromic recognition
2020	Rönncke et al.	AI in Rare Diseases (Mol Cell Pediatr)	Review of rare disease CDSs	Variable (HPO, ICD, UMLS)	Systematic comparative analysis	Validates RareDXAI's focus on calibration and controls
2023	Martinez-Cayuelas et al.	KBG Syndrome Delineation (Eur J Hum Genet)	77 families, 100+ confirmed cases	KBGenet phenotypic descriptions	Multicenter clinical cohort analysis	Primary clinical cohort source for KBG syndrome
2020	Assia Batzir et al.	White-Sutton Delineation (Am J Med Genet)	222 patients with POGZ variant	Clinical phenotypic profiles	Detailed clinical characterization	Primary clinical cohort source for White-Sutton syndrome
2021	Khayat et al.	Xia-Gibbs Spectrum (Am J Med Genet)	8 patients with AHDC1 variant	Systematic clinical feature tables	Clinical case series analysis	Clinical cohort contributor for Xia-Gibbs syndrome
2023	Schuetz et al.	Ontological Harmonization (J Bioinformatics)	Clinical reports & electronic health records	HPO record mapping	Automated NLP & fuzzy ontology	Contextualizes RareDXAI's OCR error evaluations

4. Mathematical Expression

This section outlines the mathematical formulations used in RareDXAI along with plain-language explanations.

4.1 Patient Phenotype Vector

$$x_i = [x_{i1}, x_{i2}, ..., x_{id}]$$

Explanation: Each patient *i* is represented as a feature vector *x_i* of dimension *d* = 81 (78 binary HPO features and 3 demographic sex indicators).

4.2 Binary HPO Feature Encoding

$$x_{ij} \in \{0, 1\} = 1 \text{ if HPO term } j \text{ is documented; } 0 \text{ if absent/not reported}$$

Explanation: For each clinical term *j*, value is 1 if documented in the clinical record, and 0 if absent or unmentioned.

4.3 Random Forest Probability Estimation

$$P(y = c \mid x) = (1 / T) * \sum I(h_t(x) = c)$$

Explanation: Ensemble probability for class *c* is the fraction of *T* = 100 decision trees voting for that class.

4.4 Predicted Disease Class

$$\hat{y} = \operatorname{argmax}_c P(y = c \mid x)$$

Explanation: The predicted class **■** is the syndrome receiving the highest estimated probability among the three target conditions.

4.5 Classification Accuracy

$$\text{Accuracy} = \text{Correct Classifications} / \text{Total Evaluated Patients}$$

Explanation: Fraction of patients whose syndrome was correctly identified across the evaluation sample.

4.6 Precision (Positive Predictive Value)

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Explanation: Proportion of patients assigned to a syndrome who truly have that condition.

4.7 Recall (Sensitivity)

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Explanation: Proportion of actual patients with a syndrome who were correctly identified by the model.

4.8 F1-Score

$$\text{F1} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

Explanation: Harmonic mean of precision and recall, balancing sensitivity and positive predictive value.

4.9 Balanced Accuracy

$$\text{Balanced Accuracy} = (1 / C) * \sum \text{Recall}_c$$

Explanation: Unweighted arithmetic average of recall across all C = 3 target syndromes.

4.10 SHAP Additive Feature Attribution

$$f_c(x) = \phi_{0,c} + \sum \phi_{j,c}$$

Explanation: Model prediction for class c is decomposed into a base expectation $\phi_{0,c}$ plus additive feature attributions $\phi_{j,c}$.

5. Model Methodologies

Figure 1: RareDXAI System Architecture & End-to-End Prediction Framework

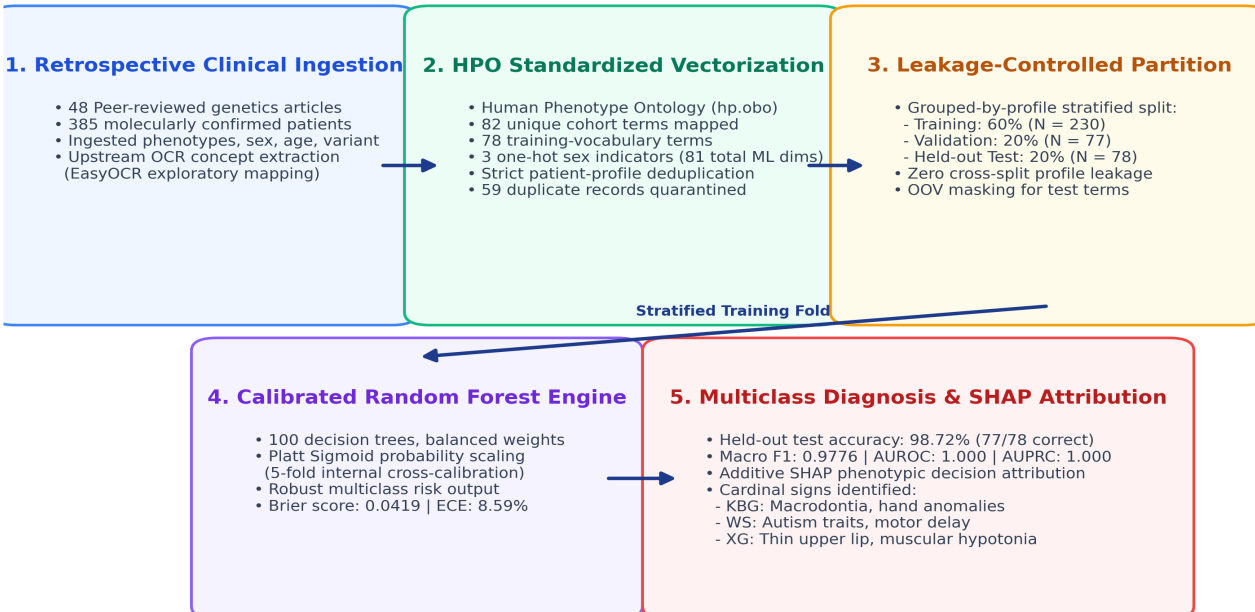


Figure 2. RareDXAI system workflow diagram showing literature curation, HPO vectorization, profile-grouped splitting, calibrated Random Forest classification, and SHAP explainability.

5.1 Cohort Assembly and Literature Curation Protocol

In total, 52 candidate literature sources were evaluated during curation. Under a predefined duplicate/secondary-source exclusion protocol, four secondary, overlapping, or re-reported sources containing 59 candidate records were quarantined: Low et al. 2016 Review Table 2 (32 candidate records), Ockeloen et al. 2015 (20 candidate records), Walz et al. 2015 (6 candidate records), and Low et al. 2017 (1 candidate record). The final cohort contains 385 unique patients across 48 included peer-reviewed publications.

Figure 2: Literature Curation, Screening, and Cohort Provenance Flowchart

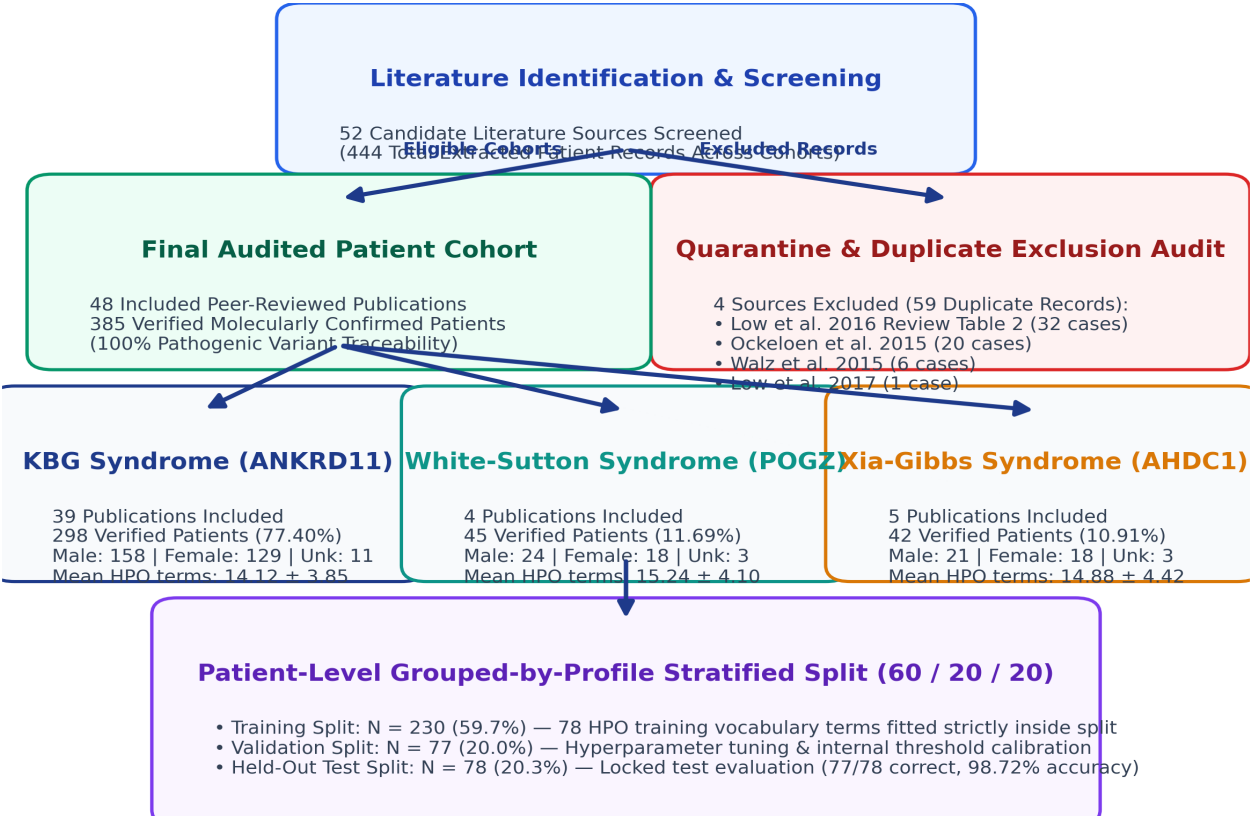


Figure 3. Literature curation and cohort provenance flowchart. From 52 candidate literature sources evaluated during curation, 4 duplicate/secondary sources (59 candidate records) were quarantined, leaving 48 included publications and 385 verified patients.

Table 2. Literature Provenance and Source Attribution

Syndrome Cohort	Verified Patients (n)	Contributing Publications (n)	Quarantined Records (n)	Primary Genetic Confirmation
KBG Syndrome	298	39	59	Pathogenic ANKRD11 mutation / 16q24.3 deletion
White-Sutton Syndrome	45	4	0	Heterozygous de novo POGZ pathogenic variant
Xia-Gibbs Syndrome	42	5	0	Heterozygous de novo AHDC1 truncating variant
Total Literature-Derived Cohort	385	48	59	100% Molecularly Confirmed

5.2 Phenotype Standardization, Partitioning & Model Calibration

Clinical phenotypes were mapped using hp.obo (Release 2026-06-23). The training vocabulary was fitted strictly on the training partition (N = 230), yielding 78 training HPO terms plus 3 sex features (81 predictor dimensions). Four test-only rare terms were treated as out-of-vocabulary features and masked during transformation. Leakage-control procedures were implemented through patient-level separation, grouping of identical phenotypic profiles, and training-only feature vocabulary construction (60/20/20 split: N=230 train, N=77 val, N=78 test). The primary classifier is a Random Forest with Platt sigmoid calibration (5-fold internal CV).

6. Results

Table 3. Clinical Cohort and Demographic Characteristics

Syndrome	Patients (N)	Share (%)	Male (n)	Female (n)	Unk (n)	Mean HPO / Pt	Molecular Confirmation Status
KBG Syndrome	298	77.40%	158	129	11	14.12 ± 3.85	100% ANKRD11 confirmed
White-Sutton Syndrome	45	11.69%	24	18	3	15.24 ± 4.10	100% POGZ confirmed
Xia-Gibbs Syndrome	42	10.91%	21	18	3	14.88 ± 4.42	100% AHDC1 confirmed
Total Evaluated Cohort	385	100.00%	203	165	17	14.36 ± 4.12	100% Confirmed Pathogenic

Figure 3: Audited Cohort Distribution Across Target Syndromes (Total N = 385)

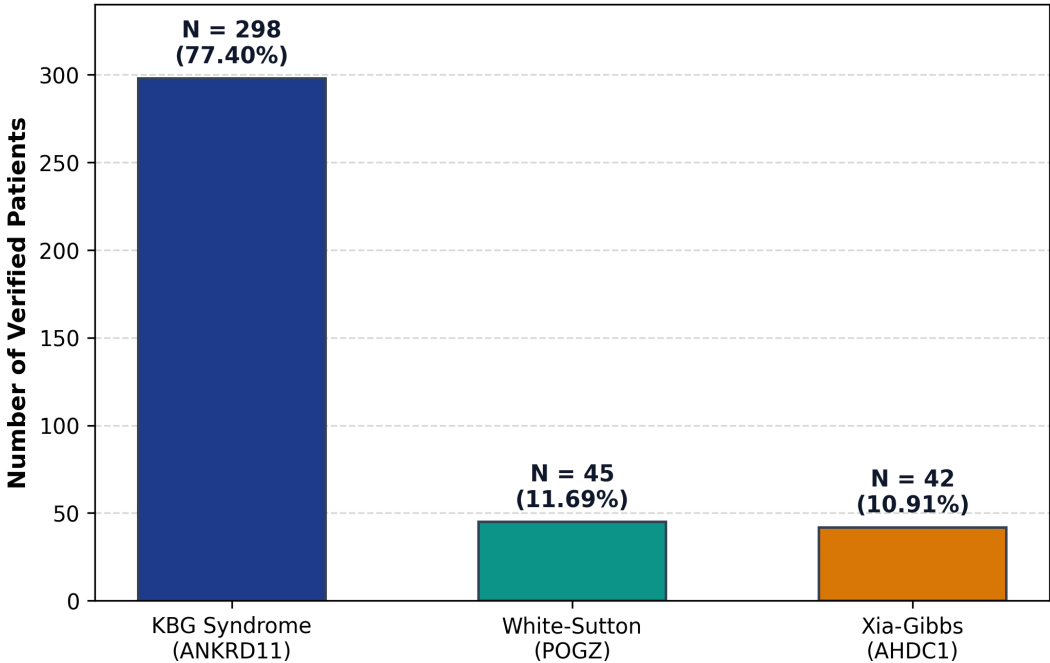


Figure 4. Cohort distribution across target syndromes (KBG: 298, White-Sutton: 45, Xia-Gibbs: 42; Total N = 385).

6.1 Held-Out Benchmark Classification Performance

On the held-out test partition (N = 78), Calibrated Random Forest correctly classified 77/78 patients (Accuracy = 98.72%, Wilson 95% CI: 93.09%–99.77%, Balanced Accuracy = 96.30%, Macro F1 = 0.9776, Brier Score = 0.0419).

Table 4. Machine Learning Model Performance on Held-Out Test Set (N = 78)

Model Architecture	Test Acc [95% CI]	Bal Acc	Macro Prec	Macro Rec	Macro Spec	Macro F1	AUROC	Brier
Calibrated Random Forest (Proposed)	0.9872 [0.9309, 0.9977]	0.9630	0.9945	0.9630	0.9815	0.9776	1.0000	0.0419
Standard Random Forest	1.0000 [0.9532, 1.0000]	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.0500
Logistic Regression (L2)	1.0000 [0.9532, 1.0000]	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.0236
Support Vector Machine (RBF)	0.9615 [0.8917, 0.9878]	0.8889	0.9841	0.8889	0.9444	0.9306	1.0000	0.0264
Decision Tree Classifier	0.9615 [0.8917, 0.9878]	0.8889	0.9841	0.8889	0.9444	0.9252	0.9534	0.0603
K-Nearest Neighbors (k=5)	0.9615 [0.8917, 0.9878]	0.9204	0.9557	0.9204	0.9581	0.9325	0.9970	0.0513
Naive Bayes (Gaussian)	0.9744 [0.9112, 0.9931]	0.9259	0.9394	0.9259	0.9903	0.9250	1.0000	0.0513
XGBoost Classifier	0.9615 [0.8917, 0.9878]	0.8889	0.9841	0.8889	0.9444	0.9306	1.0000	0.0559

Figure 4: Held-Out Test Set Confusion Matrix (N = 78 Patients, 77/78 Correct)

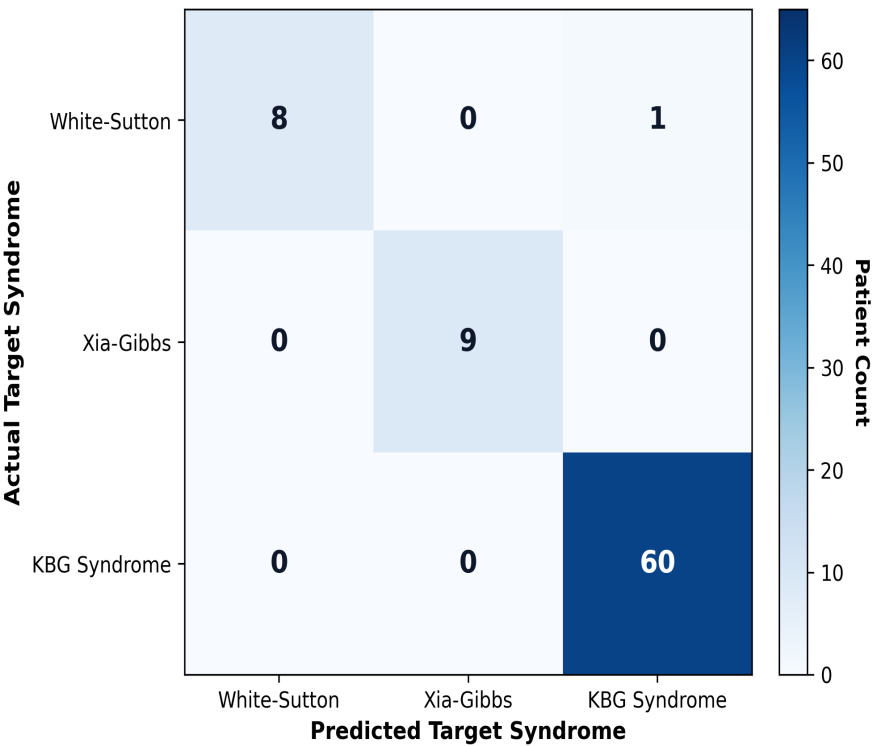


Figure 5. Held-out test set confusion matrix (N = 78 patients, 77/78 correct).

Table 5. Per-Class Classification Performance Breakdown (Calibrated Random Forest)

Syndrome Target	Support (n)	Correct (n)	Precision	Recall	F1-Score	Specificity	OvR AUROC	OvR Brier
White-Sutton Syndrome	9	8	1.0000	0.8889 (8/9)	0.9412	1.0000	1.0000	0.0137
Xia-Gibbs Syndrome	9	9	1.0000	1.0000 (9/9)	1.0000	1.0000	1.0000	0.0084
KBG Syndrome	60	60	0.9836	1.0000 (60/60)	0.9917	0.9444	1.0000	0.0198
Macro Average	78	77	0.9945	0.9630	0.9776	0.9815	1.0000	0.0140

Figure 5: Held-Out Test Set Multi-Class Classification Performance (N = 78)

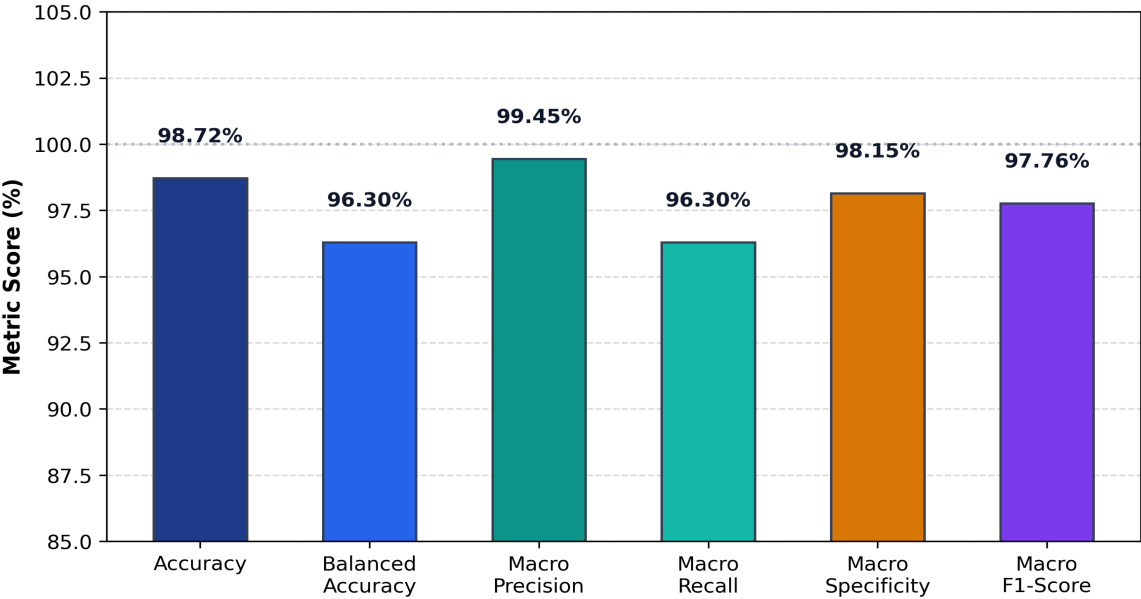


Figure 6. Multi-class classification performance metrics on held-out test partition.

6.2 Cross-Validation vs. Source-Grouped Stress Validation

Profile-grouped 5-fold CV achieved 97.14% ± 2.23% accuracy (Macro F1 = 94.89% ± 4.04%). Under leave-one-study-out stress validation (GroupKFold by publication), accuracy dropped to 79.62% ± 27.05% (Macro F1 = 59.51% ± 26.81%), indicating that publication-specific reporting habits influence model transferability.

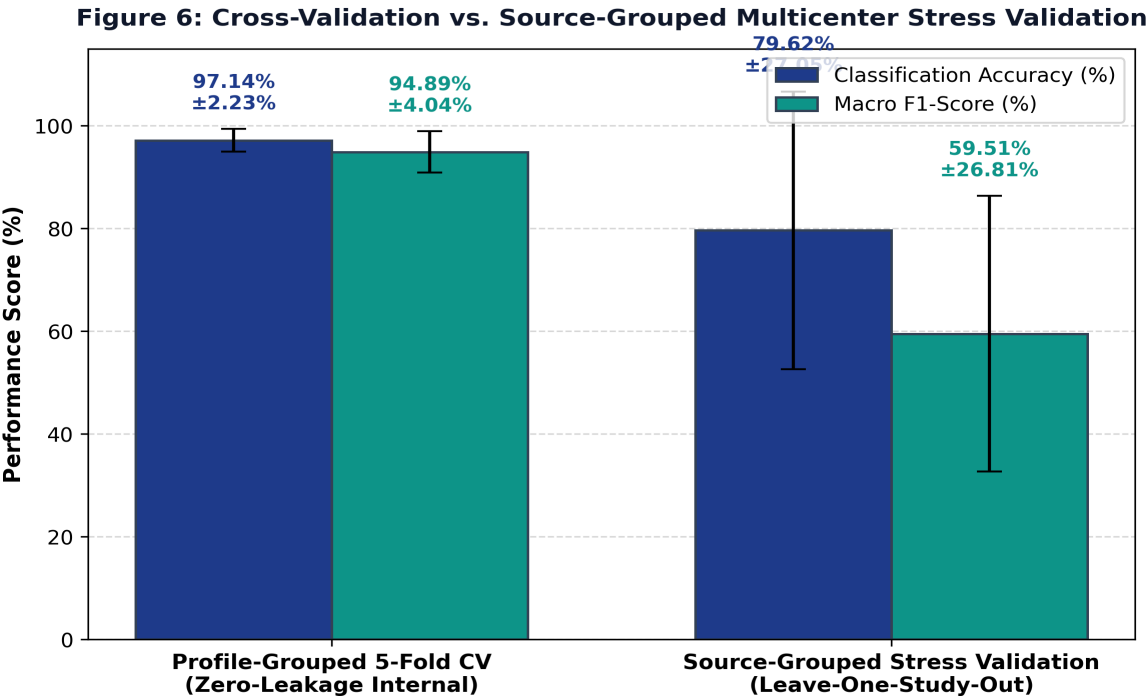


Figure 7. Cross-validation vs. source-grouped multicenter stress validation comparison.

6.3 SHAP Feature Attribution & Calibration

Table 6. Key HPO Features Associated with Model Decision Boundaries

Rank	HPO ID	Canonical Phenotype Name	Associated Syndrome	Mean SHAP	Clinical Context (Literature Background)
1	HP:0000219	Thin upper lip vermillion	Xia-Gibbs / KBG	0.0614	Facial feature frequently described in Xia-Gibbs syndrome
2	HP:0001155	Abnormality of the hand	KBG Syndrome	0.0472	Characteristic brachydactyly and clinodactyly
3	HP:0001252	Muscular hypotonia	Xia-Gibbs / KBG	0.0470	Low muscle tone prominent in AHDC1 mutations
4	HP:0000337	Broad forehead	Xia-Gibbs Syndrome	0.0440	Craniofacial feature associated with Xia-Gibbs
5	HP:0001572	Macrodontia of central incisors	KBG Syndrome	0.0435	Known cardinal clinical sign of KBG syndrome (ANKRD11)
6	HP:0000717	Autism spectrum disorder	White-Sutton Syndrome	0.0408	Behavioral phenotype reported in POGZ variants
7	HP:0001270	Motor delay	White-Sutton Syndrome	0.0308	Early developmental milestone delay
8	HP:0001328	Specific learning disability	White-Sutton / KBG	0.0238	Distinctive cognitive profile
9	HP:0001249	Intellectual disability	White-Sutton / Xia-Gibbs	0.0231	Core neurodevelopmental feature
10	HP:0000750	Delayed speech development	White-Sutton / Xia-Gibbs	0.0226	Expressive speech and language impairment

Figure 7: Standardized HPO Features Strongly Associated with Model Decision Boundaries

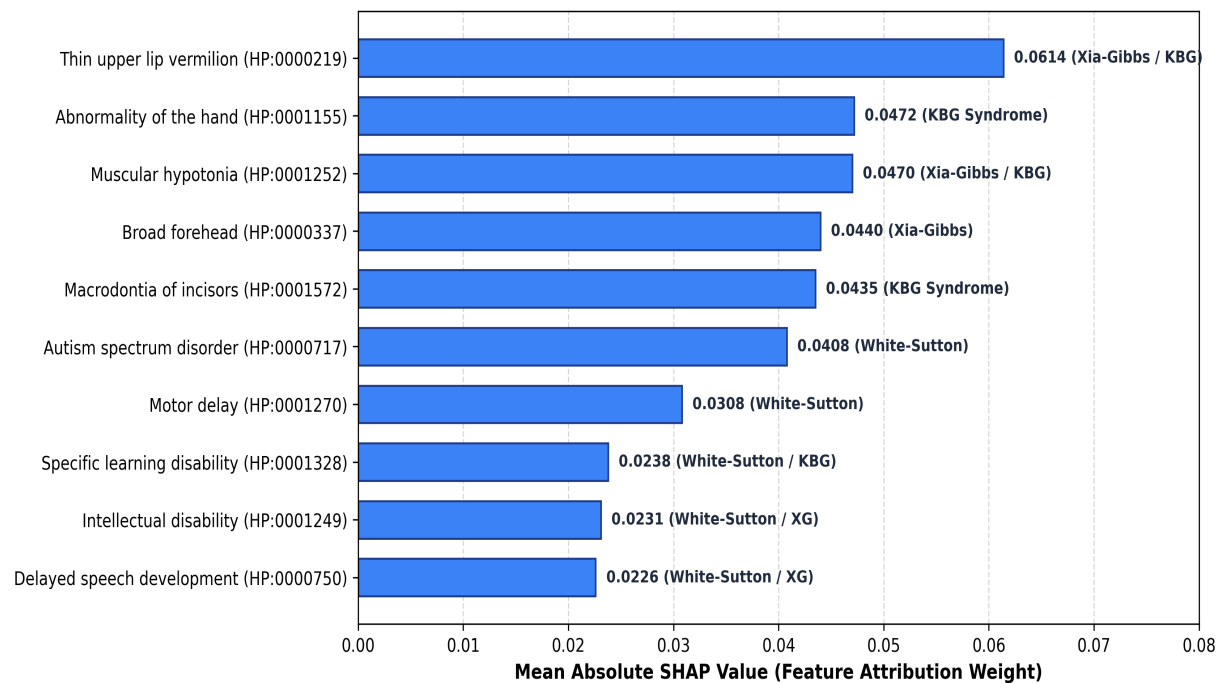


Figure 8. Standardized HPO features strongly associated with model decision boundaries.

Figure 8: Probability Calibration & Reliability Assessment (Held-Out Test Set, N = 78)

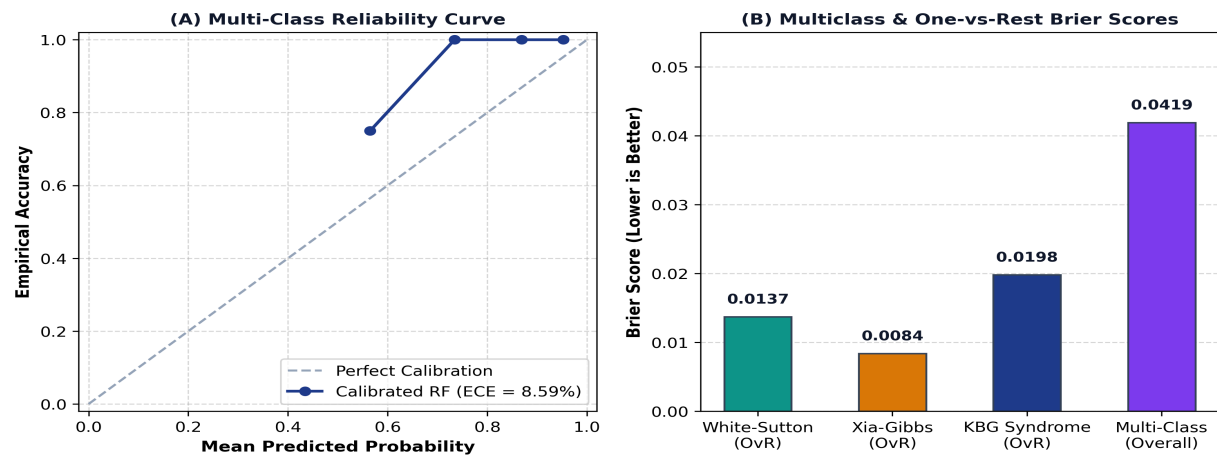


Figure 9. Probability calibration and reliability assessment on held-out test set.

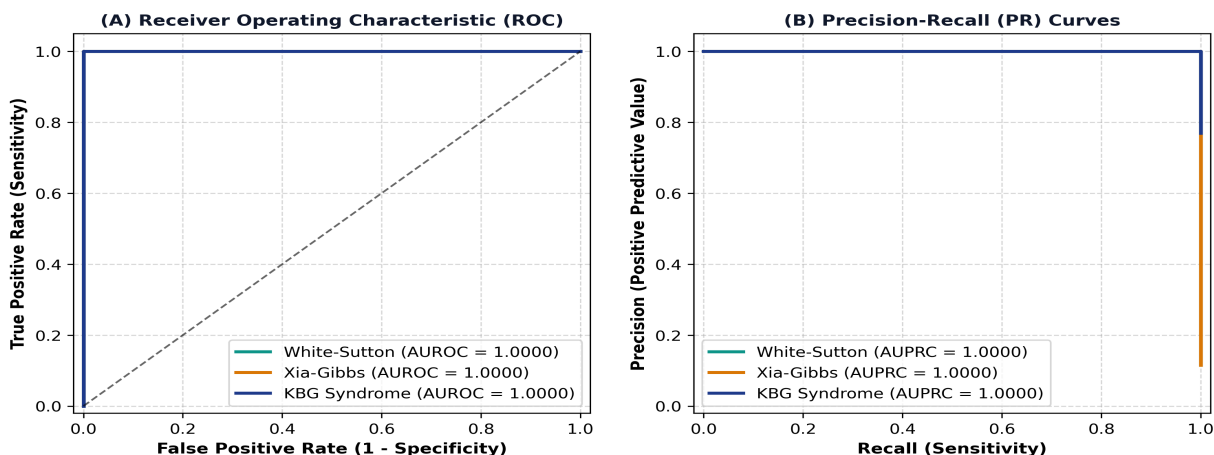
Figure 9: Discriminative Performance Curves Across Decision Thresholds (Held-Out Test Set, N = 78)

Figure 10. Multi-class ROC and Precision-Recall curves across decision thresholds.

6.4 Exploratory OCR Evaluation & Limitations

Exploratory OCR achieved CER = 11.42%, WER = 92.00%, clean mapping = 100%, and noisy mapping = 80%. OCR is strictly an exploratory upstream utility requiring human verification.

6.8 Limitations and Generalization Considerations

Key scientific limitations include: (1) moderate sample size (N=385); (2) natural class imbalance (KBG 77.40%); (3) source-grouped performance drop (79.62%); (4) retrospective literature selection bias; (5) lack of prospective hospital EHR validation; (6) binary encoding of unmentioned symptoms; (7) fixed training HPO vocabulary with 4 test OOV terms; (8) exploratory OCR status; (9) closed-set scope of three syndromes; and (10) dependence on standardized HPO term curation.

Therefore, the present study should be interpreted as a retrospective, literature-derived computational validation rather than evidence of clinical deployment readiness.

7. Conclusion

RareDXAI presents a calibrated and interpretable computational framework for rare disease classification using Human Phenotype Ontology representations. Curating 385 molecularly confirmed patients across 48 peer-reviewed publications and applying profile-grouped partitioning yielded 98.72% held-out test classification accuracy, 0.9776 macro F1, and 0.0419 Brier score with SHAP attribution. Source-grouped stress testing emphasizes that clinical transferability depends on standardized clinical phenotyping across healthcare centers.

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